INCH-POUND
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# DETAIL SPECIFICATION

# CARRIERS, FULL-TRACKED: M548A1 AND M730A1

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two types of unarmored full-tracked, diesel powered vehicles. The M730A1 vehicle is designed for cross country and amphibious operation as a guided missile carrier. The M548A1 vehicle is for cross country operation only, as cargo and ammunition support carrier (see 6.1).

1.2 Classification. Vehicles are one of the following models (see 6.2).

| Model | Army drawing | Vehicle type |
|---|---|---|
| M548A1 | 8750029 | Carrier, Cargo Full-tracked |
| M730A1 | 8750067 | Carrier, Guided Missile Equipment |

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2350

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| A-A-50271 | - Plate, Identification. |
| A-A-52557 | - Fuel Oil, Diesel; For Posts, Camps, and Stations. |

DEPARTMENT OF DEFENSE

| | |
|---|---|
| MIL-PRF-62099 | - Transmission, Hydraulic: Modified Automatic TX-100-1. |
| MIL-PRF-62140 | - Engine, Diesel: Liquid Cooled, V-Type, 6 Cylinder, 210 H.P. |

STANDARDS

DEPARTMENT OF DEFENSE

| | |
|---|---|
| MIL-STD-461 | - Requirements for the Control of Electromagnetic Interference Emission and Susceptibility. |
| MIL-STD-462 | - Measurement of Electromagnetic Interference Characteristics. |
| MIL-STD-642 | - Identification Markings of Combat and Tactical Transport Vehicles. |
| MIL-STD-1275 | - Electrical Circuit, 28 Volt DC, Transient Characteristics for Military Vehicles. |

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.



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NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DRAWINGS

| | |
|---|---|
| 8750029 | - Carrier, Cargo, M584A1. |
| 8750067 | - Carrier, Guided Missile Equipment, M730A1. |
| 10942803 | - Lubrication Chart, Production, Fuel and Coolant. |
| 12369001 | - Aluminum External Surfaces, Green 383. |
| 12369007 | - Aluminum Internal Surfaces, Light Green. |

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

| | |
|---|---|
| ANSI Z1.4 | - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted). |

(Application for copies may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| ASTM B117 | - Standard Practice for Operating Salt Spray (Fog) Apparatus (DoD Adopted). |
| ASTM D522 | - Standard Test Methods for Mandrel Bend Test of Attached Organic Coatings (DoD Adopted). |

(Application for copies may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)



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GENERAL MOTORS CORPORATION

GM 9540P - Accelerated Corrosion Test.

(Copies of GM publications may be obtained from General Motors Corporation, c/o Global Engineering, 15 Inverness Way, Englewood, CO 80112.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's material specifications for full-tracked carriers. The materials shall be capable of meeting all the operational and environmental requirements specified herein (see 4.5.1). Materials and special treatments should be utilized to prevent corrosion and to resist harmful effects of dust, water and fungi. Dissimilar metals shall not be used in intimate contact without protection from galvanic corrosion. Asbestos, cadmium, and radioactive material will not be used in this item. Radioactive material is defined by Title 10, Code of Federal Regulations (CFR), Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries.

3.2.1 Corrosion protection. The vehicle shall be fabricated from compatible materials providing corrosion protection and coating adherence equal to or exceeding that provided by hot dip galvanized 1010 steel with a minimum coating thickness of 2 mils (50 micrometers) on steel sheet less than 0.0625 inches (in.) thick, or 0.75 mil (19 micrometers) on pre-galvanized steel sheet 0.0625 in. thick or less. A material sample shall be capable of meeting or exceeding the corrosion resistance provided by a galvanized sample (as described above) when subjected to the Mandrel Bend Test of ASTM D522, and followed by the Accelerated Corrosion Test of GM 9540P Method B, 120 cycles (see 6.2).

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.



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3.3 Construction. The vehicle shall be constructed in accordance with the following drawings, as applicable, and applicable specifications and standards (see 4.5.2). If more than one unit of a particular component is used, the components used shall be identical in make, material and quality shall be the same on each vehicle. Completed vehicles shall meet all requirements specified herein, and be of a quality that assures a vehicle free of defects that would compromise, limit, or reduce performance.

| Model | Drawing number |
|---|---|
| M588A1 | 8750029 |
| M730A1 | 8750067 |

3.3.1 Bearing seals. When fording, operating amphibiously (M730A1 only), or in mud, sand or snow, bearing seals shall prevent the entrance of foreign matter into the bearings, which are exposed to contamination during these operations. Bearing seals shall restrict the leakage of lubricants from bearings (see 4.5.5).

3.3.2 Special kits. Provisions shall be made for installation of special kits including winterization equipment. When specified (see 6.2), special kits shall be furnished as equipment. Special kits, when installed, shall fit properly without interference or misalinement and meet applicable environmental and performance requirements specified herein (see 4.5.6).

3.3.2.1 Winterization kit (M730A1). When installed (see 6.2), the winterization kit shall supplement the personnel heater by providing insulation for the vehicle cab, frost shields for the cab door windows, and an insulated cab cover assembly to permit vehicle operation in ambient temperatures from -25 degrees Fahrenheit (°F) to -65°F. The insulated soft cab cover shall resist the entrance of water, dust and snow. When installed, the engine coolant heater system shall assure engine starting and operation in ambient air temperature from -25°F to -65°F. The engine coolant heater shall be capable of operating for a minimum of twelve hours on battery power alone without discharging the battery to a level that would preclude satisfactory starting of the engine (see 4.5.6.1).

3.3.2.2 Cab personnel heater kit (primary) (M548A1). When installed, the (primary) cab personnel heater kit shall provide heat for personnel and windshield defrosting in ambient temperatures down to -25°F (see 4.5.6.2).

3.3.2.3 Cab personnel heater kit (secondary) (M548A1). When installed, the (secondary) cab personnel heater kit shall supplement the (primary) cab personnel heater and ensure heat for personnel and windshield defrosting during operations in ambient temperatures from -25°F to -65°F. The insulation hard top cab cover shall resist the entrant of water, dust and snow (see 4.5.6.3).



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3.3.2.4 Cargo area personnel heater kit (primary) (M548A1). When installed, the (primary) cargo area personnel heater kit shall provide heat for personnel in the rear cargo area in ambient temperatures down to -25°F (see 4.5.6.4).

3.3.2.5 Cargo area personnel heater kit (secondary) (M548A1). When installed, the (secondary) cargo area personnel heater kit shall supplement the (primary) cargo area personnel heater kit during operations in ambient temperatures from -25°F to -65°F. The insulated cargo cover shall provide protection for personnel and material in the cargo area in ambient air temperatures down to -65°F. The insulated cargo cover shall be water, dust and snow proof (see 4.5.6.5).

3.3.2.6 Engine coolant heater kit (M548A1). When installed, the engine coolant heater kit shall assure engine starting and operation in ambient air temperatures from -25°F to -65°F. The engine coolant heater shall be capable of operating for a minimum of twelve hours on battery power alone without discharging the battery to a level that would prevent satisfactory starting of the engine (see 4.5.6.6).

3.3.2.7 Air brake kit (M548A1). When installed, the air brake kit shall adequately function with air brakes on applicable vehicles or equipment. NOTE: The air brake kit is required on carriers that are to be used as prime movers of vehicles or of equipment that has air brakes (see 4.5.6.7).

3.3.2.8 Material handling kit (M548A1). When installed, the material handling kit shall be capable of loading material weighing up to 1500 pounds (lb) into the cargo area. This kit shall be capable of being mounted either left of center, right of center, or in a center location of the cargo area, without interference (see 4.5.6.8).

3.3.2.9 Machine gun kit (M548A1). When mounted on vehicle, the machine gun kit shall be capable of supporting the machine gun when firing, and allow the gun to rotate 360° to the right and left without interference or binding (see 4.5.6.9).

3.3.3 Doors, windows, covers, and miscellaneous chassis items. All doors, windows, covers, hatches, vision devices, receptacles, ducts, vents, panels, stowage boxes, seats, locking devices and fenders shall conform to applicable drawings. When installed, they shall fit and function without interference. When shut, all doors, windows, hatches, and covers shall be water, dust, and snow resistant (see 4.5.2 and 4.5.7).

3.3.4 Engine. The engine shall conform to MIL-PRF-62140 and shall operate throughout all speed ranges without evidence of cooling system, lubricant, or fuel leakage (see 4.5.2 and 4.5.8).



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3.3.4.1 Engine air induction system. The air induction system, as installed, shall prevent entrance of foreign matter in conformance with applicable drawings (see 4.5.8.1).

3.3.5 Exhaust systems. The engine exhaust system, as installed, and the personnel heater exhaust system, when installed, shall not permit concentration of toxic fumes in the driver and crew compartments to exceed 50 parts per million (ppm) for carbon monoxide (CO) or 5 ppm for oxides of nitrogen (NO) (see 4.5.9).

3.3.6 Fuel system. The fuel system shall maintain a continuous supply of fuel to the engine when ascending and descending 60% grades in forward and reverse gear, and when vehicle is being operated left side or right side up on 30% side slope (see 4.5.10).

3.3.6.1 Fuel tank. Provision shall be made to assure the internal cleanliness of the fuel tank prior to initial fueling of vehicle. The fuel tank shall be capable of receiving 50 gallons (gal) of fuel per minute (see 4.5.10.1).

3.3.6.2 Fuel lines. Fuel lines and connections shall be internally clean prior to initial fueling and servicing of vehicle. All fuel lines, connections, and tanks shall be free of leaks (see 4.5.2 and 4.5.10.2).

3.3.6.3 Rack linkage. With rack linkage installed in accordance with applicable drawings, and with the accelerator pushed against the stop, the rack shall be at full open position (see 4.5.2 and 4.5.10.3).

3.3.7 Transmission, transfer gearcase, and control differential. The transmission shall conform to MIL-PRF-62099. When installed in vehicle, the transmission and controlled differential shall operate throughout all gear ranges without evidence of leaking (see 4.5.2 and 4.5.11).

3.3.8 Electrical system.

3.3.8.1 Generating system voltage (alternator). After no less than 20 minutes of operation with headlights on, regulator control shall be observed. Alternator shall be operated at 4000 revolutions per minute (rpm) $\pm$100 rpm with an output of 50 $\pm$5 amperes, and the voltage regulator rheostat adjusted to provide 28 volts direct current (Vdc) at the vehicle batteries. Voltage at the vehicle batteries shall not increase more than 0.5 Vdc when the headlight load is removed (see 4.5.12.1).

3.3.8.2 Lights. Internal and external lights specified on applicable drawings shall operate, and all electrical contacts and connections shall maintain positive contact, under all vehicle operating conditions (see 4.5.2 and 4.5.12.2).



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3.3.8.3 Communication and electrical circuits. Communication and electrical circuits shall maintain continuity from end to end, without shorts from pin to pin, or from pin to connector shell (see 4.5.2 and 4.5.12.3).

3.3.8.4 System voltage characteristics. Electrical system voltage characteristics shall fall within the limits for transient and steady state voltage requirements specified in MIL-STD-1275 (see 4.5.12.4).

3.3.8.5 Electromagnetic interference suppression. The vehicle, when operating, shall meet the requirements for EMI/EMC suppression in Army ground vehicles as specified in MIL-STD-461 (see 4.5.12.5).

3.3.8.6 Heaters. The personnel and cargo area heaters shall start and operate satisfactorily on 20 Vdc, using diesel fuel conforming to A-A-52557 (see 4.5.2 and 4.5.12.6).

3.3.9 Controls. All electrical, mechanical and hydraulic controls shall operate without malfunction throughout all ranges of operation under all vehicle operating conditions (see 4.5.2 and 4.5.13).

3.3.10 Adjustment mechanisms. All adjustment mechanisms shall function properly and maintain adjustment within required settings during all vehicle operating conditions (see 4.5.2 and 4.5.14).

3.3.11 Stowed material. All interior and exterior on vehicle material (OVM), when stowed on the vehicle in the spaces provided, shall fit properly and not interfere with the operation of the vehicle and components in any manner (see 4.5.15).

3.3.12 Folding blast covers (M730A1). The folding blast covers shall provide protection for the crew compartment when launching a missile. The blast covers shall fold open and remain open to permit vehicle operation without cover closing on operator (see 4.5.16).

3.3.13 Tailgate (M730A1). The tailgate shall hinge downward and support the weight of two men during manual missile loading of the launcher. When closed the tailgate shall be sealed to restrict the entrance of water into the vehicle during fording and flotation operations (see 4.5.17).

3.3.14 Air transportability (M730A1). The vehicle when mated with its missile pallet shall be capable of Phase II transport in a C-130 aircraft (see 4.5.18).

3.4 Break-in run. Unless otherwise specified (see 6.2), prior to being subjected to testing, each vehicle shall be given a break-in run (see 4.5.3). Break-in is not intended to be a part of any test but is carried out by the contractor to ensure that the vehicle is in a safe, operable condition



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prior to testing. A combat-loaded condition is not required. When specified (see 6.2), the break-in run shall be simulated (see 4.5.3.3).

3.5 Performance. A complete vehicle, loaded with rated payload, serviced as specified in 4.3, and equipped for existing climatic conditions, shall operate as specified herein without special equipment. Performance shall be demonstrated on smooth, level, hard-surfaced roads free of loose material, except that a section of road shall be as shown in figure 1.

3.5.1 Reliability. Reliability shall be not less than 750 mean miles between failure (MMFB) when subjected to a 5000 mile operational test (see 4.5.4).

3.5.2 Durability. The vehicle with rated payload shall have a 50% probability of completing the 5000 test miles (see 4.5.4) without replacement, rebuild, or major overhaul of the engine, transmission, transfer gear unit, differential steer unit, or final drives, and without repair or rebuild of the hull.

3.5.3 Maintainability. Total maintenance, excluding driver/crew checks and services, shall not exceed 175 manhours during the 5000 mile operational test (see 4.5.4).

3.5.4 Environmental conditions. The complete vehicle with rated payload shall operate in ambient air temperature of +125°F to -25°F and, with special equipment installed, to -65°F. The complete vehicle shall withstand storage at -80°F without deterioration that may cause failure of any component part of the vehicle (see 4.5.19).

3.5.5 Cooling systems.

3.5.5.1 Engine cooling. With the vehicle operating in an ambient temperature of 125°F, the engine cooling system shall maintain engine coolant at not more than 230°F measured at a point in the engine outlet to radiator. The radiator pressure cap shall relieve between 13 and 18 pounds per square inch gage (psig). Air conditioning or microclimatic cooling should be used to provide cooling for the operators of these vehicles to allow them to effectively operate the vehicles at their mission parameters (see 4.5.2 and 4.5.20).

3.5.5.2 Transmission, lubricant cooling. With the vehicle operating in an ambient temperature of 125°F, the transmission lubricant cooling system shall maintain lubricant temperature at not more than 325°F, at the point where the coolant flows out of the radiator and into the cooler except at torque converter speed ratios less than four to ten (see 4.5.2 and 4.5.20).

3.5.5.3 Differential oil cooler. With the vehicle operating in an ambient temperature of 125°F, the differential lubricant cooler shall maintain lubricant temperature below 300°F, measured out of differential into cooler (see 4.5.2 and 4.5.20).



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3.5.6 Bilge pumps. The bilge pump(s), installed in accordance with applicable drawings, shall remove water from the vehicle at a minimum rate of 35 gallons per minute (gal/min) each with a discharge head of 4-1/2 feet (ft), and with pump motor operating on 27.5 Vdc impressed at the motor (see 4.5.2 and 4.5.21).

3.5.7 Speeds and vehicle control. The vehicle shall sustain speeds up to 32 mph, and a maximum speed of 38 mph after 400 miles of operation, without damage to power plant or suspension system. With vehicle traveling between 15 mph and maximum speed, drift shall not exceed 3 ft in 100 ft without turning the steering wheel (see 4.5.22).

3.5.8 Acceleration. With the vehicle at a standstill, transmission in the 1-3 range and engine at idle speed, the vehicle shall accelerate from 0 to 20 miles per hour (mph) in not more than 18 seconds with rated payload or in not more than 11 seconds without load on a smooth, level, hard-surfaced road (see 4.5.23).

3.5.9 Engine starting. With the vehicle parked in the various positions listed below, the engine shall be capable of restarting within 1 minute after the following cycle: operation at 750 to 1000 rpm under no load for 2 minutes and thereafter stopped for 1 minute minimum (see 4.5.24):

a. Headed upgrade on 60% grade.
b. Headed downgrade on 60% grade.
c. Left side up on 30% slope.
d. Right side up on 30% slope.

3.5.10 Braking.

3.5.10.1 Stopping and vehicle control. Vehicle shall stop within a distance of 40 ft from point of service brake application when traveling at a speed of 20 mph on smooth, level, hard-surfaced roads free of loose material. Drift shall not exceed 3 ft in stopping distance. During stopping, the vehicle being under driver's individual control, steering action is permitted (see 4.5.25).

3.5.10.2 Holding. With brakes locked for parking and with transmission in neutral, the vehicle shall hold stationary when parked on a 60% longitudinal grade. For this operation vehicle shall be loaded with rated payload or towing a trailer (see 4.5.26).

3.5.11 Turning. Vehicle, on smooth, level, hard-surfaced road or track, with differential steer, shall make continuous, full 360° turns to right and to left, with rear point of outer edge of vehicle at all times in contact with or within a 50 ft diameter circle. When in pivot steer, the vehicle shall pivot within a 28 ft diameter circle. The applied effort to the steering lever shall not exceed 100 lb to pivot the vehicle (see 4.5.27).



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3.5.12 Water operation.

3.5.12.1 Shallow water fording. The vehicle shall ford shallow, hard-bottom salt or fresh water channels up to a depth of 40 inches (in.). Leakage of water into the hull shall not exceed 3 quarts per minute (qt/min) (see 4.5.28).

3.5.12.2 Amphibious operation (M730A1). With flotation kit installed, the M730A1 vehicle shall negotiate water of unlimited depth with height of waves not exceeding 1 ft. When afloat in calm water not less than 15 ft deep, a maximum speed of not less than 3 mph shall be attained. The vehicle shall be capable of operating within a 30 ft corridor when crossing calm, open water. Leakage of water into the hull shall not exceed 3 qt/min (see 4.5.29).

3.5.13 Trench crossing. Vehicle shall cross trenches up to 36 in. in depth, and up to 66 in. in width, without damage to vehicle (see 4.5.30).

3.5.14 Vertical obstacles. Vehicle, operating at low speeds in forward gear, shall cross vertical obstacles (wall) up to 24 in. in height and, in reverse gear, obstacles up to 14 in. in height without damage to vehicle (see 4.5.31).

3.5.15 Grade and slope operation. Vehicle shall ascend 60% grades in forward and reverse gear, from a standstill position on the grade, without stalling or damage to power plant or power train. Vehicle shall operate without malfunction or damage on 30% side slopes, with right and then left side of vehicle up-slope. Proper oil pressure and temperature shall be maintained in the engine, transmission, transfer gear unit, and controlled differential during all grade and slope operation. Oil pressure observation shall be made with engine operating at 600 to 800 rpm (see 4.5.32).

3.5.16 Suspension. The vehicle suspension shall operate over a minimum of three 6 x 6 in. obstacles, spaced 30 ft apart, at a speed of 18 to 22 mph, without bottoming out of the track drive sprockets, roadarm bump stops, or idler wheels, with either side of both tracks, engaging the obstacles (see 4.5.2 and 4.5.33).

3.5.17 Winch system. The winch shall withstand an increasing load of more than 14 000 lb when tested as specified in 4.5.34.

3.5.17.1 Winch operation. The winch shall completely and evenly reel in the entire cable (see 4.5.34.1).

3.5.17.2 Winch structural integrity. The winch structural integrity shall be maintained when pulling the winch cable with a 20 000 lb load (see 4.5.34.2).



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3.5.17.3 Winch controls. The cab-mounted controls shall engage the power takeoff drive for operation of the winch. The manually operated device furnished shall permit free spooling of the winch drum (see 4.5.34.3).

3.6 Painting. Exterior and interior surfaces of the vehicle shall be cleaned, pre-treated, primed and painted to the following minimum performance requirement:

a. Cleaning. Surface free of surface oxides, rust, weld splatter, and other organic contaminants providing a water break-free surface.
b. Pre-treatment. Surface pre-treated within 4 hours after cleaning with chemical conversion coating or equivalent. The pre-treatment coating shall be uniform, continues and free from areas of powdery or loose coating, voids, scratches, and flaws.
c. Painting system. Primed and top-coat painted with the total dry film thickness not to exceed 5 mils.

The corrosion protection, when tested per ASTM B117 for 168 hours shall produce no more than 5 isolated spots or pits on a specimen panel size of 75 millimeters (mm) by 250 mm by 1 mm thick (as a general rule a corrosion spot or pit usually displays a characteristic tail or fine) and no more than 15 isolated spots or pits on the combined surface area of five specimen panels. None of the spots or pits shall be larger than 1 mm in diameter. Areas within the 6 mm from the edges, identification markings, and holding points shall be excluded from the evaluation. For the paint adhesion test, no intercoat separation shall occur for the paint system and the conversion coating or between the conversion coating and the base metal when two specimen test panels are tested as follows. Apply 1 in. wide pressure sensitive tape to a surface free of blemishes and make 11 parallel cross cuts 1–2 mm apart, and about 20 mm long. Within 1-2 minutes of application, rapidly remove the tape. The tape shall have an average adhesion of 60 ounces per inch (oz/in.). The exterior paint shall be forest green (see 4.5.2). When Chemical Agent Resistant Coating (CARC) paint is specified (see 6.2), the exterior surface shall be painted per CARC “A” size Drawing 12369001, and the interior surface shall be painted per CARC “A” size Drawing 12369007.

3.7 Identification marking. Vehicle marking shall be in accordance with MIL-STD-642. Identification marking shall include, as a minimum, the following information permanently affixed on a plate (see 4.5.2):

a. Part or identification number (PIN).
b. Contract or order number.
c. Manufacturer's name and trademark.
d. Date of manufacture.
e. U. S.



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3.8 Name, shipping and service data plates. Plates shall meet the requirements of A-A-50271, or applicable drawings (see 4.5.2).

3.9 Noise level. Noise levels shall not exceed 103 decibels (dB(A)) when tested according to 4.5.35. Warnings shall be added to the vehicle and the passenger compartments stating that hearing protection is required.

3.10 Fire extinguisher. The fixed fire extinguisher system, when installed in accordance with applicable drawings, shall be readily accessible for servicing and operation internally and externally. Maximum effort for externally discharging system shall not exceed value specified on applicable drawing (see 4.5.2 and 4.5.36).

3.11 Lubrication. Vehicles shall be appropriately lubricated as specified in Drawing 10942803, or in accordance with equivalent commercial practices (see 4.5.2).

3.12 Workmanship. Workmanship shall be of such quality as to assure that the vehicle and its components are free of defects that compromise, limit or reduce capability in the performance of its intended use. In addition to general appearance, the defects listed in table I shall be considered to be cause for failure (see 4.5.2).

## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on preproduction and initial production samples as specified when a first article sample is required (see 3.1).

4.2.1 Preproduction inspection. When specified (see 6.2), first article preproduction shall be performed on one vehicle. This inspection shall include the examinations of 4.4 (see table I).



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TABLE I. Classification of defects.

| Category | Defect | Method of examination |
|---|---|---|
| Major: | | |
| 101 | Construction; improper dimensions affecting performance (see 3.3). | Visual and Functional |
| 102 | Hull; structural and welding defects (see 3.3). | Visual and Functional |
| 103 | Towing pintle, lifting eyes, hooks and brackets; malfunction, defects (see 3.3). | Visual and Functional |
| 104 | Doors, windows, covers, hatches, vision devices, receptacles, ducts, vents, panels, stowage boxes, seats, locking devices, and fenders; malfunction, damage (see 3.3.3). | Visual and Functional |
| 105 | Engine; malfunction, leaks, improper mounting (see 3.3.4). | Visual and Functional |
| 106 | Fuel system components; malfunction, leaks, improper tube/hose routing, damage (see 3.3.6.2, 3.3.6.3) 1/. | Visual and Functional |
| 107 | Transfer gear, differential steer, and final drive; malfunction, leaks (see 3.3.7). | Visual and Functional |
| 108 | Transmission; malfunction, leaks, improper mounting (see 3.3.7). | Visual and Functional |
| 109 | Electrical switches, instrumentation, communication system, warning, indication and safety devices, heaters; malfunction, damage (see 3.3.8.2, 3.3.8.3, and 3.3.8.6). 2/ | Visual and Functional |
| 110 | Controls; malfunction, improper clearances (see 3.3.9). | Visual and Functional |
| 111 | Adjustment mechanism; malfunction, improper clearances (see 3.3.10). | Visual and Functional |
| 112 | Cooling system components; malfunction, leaks, damage (see 3.5.5.1, 3.5.5.2, and 3.5.5.3). | Visual and Functional |
| 113 | Bilge pumps; malfunction, damage (see 3.5.6). | Visual and Functional |
| 114 | Suspension system components and adjustments; damage, damage, leaks, improper clearances, misalignment (see 3.5.16). | Visual and Functional |
| 115 | Tracks; damage, pad separation (see 3.5.16). | Visual and Functional |
| 116 | Track, drive sprockets and drive shafts; damage, clearance improper (see 3.5.16). | Visual and SIE 3/ |



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TABLE I. Classification of defects - Continued.

| Category | Defect | Method of examination |
|---|---|---|
| 117 | Fire extinguisher system control; seals mission, components damaged (see 3.10). | Visual and Functional |
| Minor: | | |
| 201 | Construction; improper dimensions affecting appearance (see 3.3). | SIE |
| 202 | Towing pintle, lifting eyes, hooks and brackets- improper fit installation, assembly, or adhesion welding defects, damaged (see 3.3). | Visual |
| 203 | Doors, windows, covers, hatches, grilles, vision devices, receptacles, ducts, vents, panels, stowage boxes, seats, straps, crash pads, fenders-improper assembly, installation, fit, adjustments, defective welds, seals, adhesion, hardware, sheet metal (see 3.3.3). | Visual |
| 204 | Engine; improper component assembly (see 3.3.4). | Visual |
| 205 | Fuel system components-improper line clearance, assembly or installation, coding or protection, defective, missing (see 3.3.6.2). | Visual |
| 206 | Transmission, improper component (see 3.3.7). | Visual |
| 208 | Transfer gear, differential steer and final drive units; improper component assembly or installation (see 3.3.7). | Visual |
| 208 | Electrical switches, instrumentation, communication system, warning, indication and safety devices, heaters; improper assembly, installation (see 3.3.8.2, 3.3.8.3, 3.3.8.6). 2/ | Visual |
| 209 | Controls; improper assembly or installation (see 3.3.9). | Visual and SIE |
| 210 | Adjustment mechanisms; improper assembly or installation (see 3.3.10). | Visual |
| 211 | Cooling system components; improper assembly, installation, coding or protection, defective, missing (see 3.5.5.1, 3.5.5.2, 3.5.5.3). 1/ | Visual |
| 212 | Bilge pumps; improper assembly, installation (see 3.5.6). | Visual |
| 213 | Suspension system components and adjustments; improper installation or assembly, bolt or nut torque (see 3.5.16). | Visual and SIE |
| 214 | Tracks; tension, improper installation or assembly, bolt, torque, damage (see 3.5.16). | Visual and SIE |
| 215 | Track drive sprockets and drive shafts; improper assembly, installation, bolt torque (see 3.5.16). | Visual and SIE |



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TABLE I. Classification of defects - Continued.

| Category | Defect | Method of examination |
|---|---|---|
| 216 | Paint applications, color, protective anti-skid coatings; improper (see 3.6). | Visual |
| 217 | Decals, marking data and instruction plates; missing, improper location or size, defective (see 3.7 and 3.8). | Visual |
| 218 | Fire extinguisher system control; data card incomplete, improper assembly, installation, coding or protection (see 3.10). | Visual |
| 219 | Lubrication and lubricant; improper application, materials and coverage improper (see 3.11). | Visual |
| 220 | Workmanship; faulty (see 3.12). | Visual |

1/ Fuel leaks constitutes a special defect and will result in total inspection of the lot for this defect (see 6.5.2).
2/ Filament failures are not to be classified as major or minor defects.
3/ SIE = Standard Inspection Equipment.

4.2.2 Initial production inspection. Unless otherwise specified (see 6.2), first article initial production inspection shall be performed on two vehicles. This inspection shall include the examinations of 4.4 (see table I) and the tests of 4.5.4 through 4.5.36 (see table II). Subsequently, both vehicles shall be tested for 5000 miles as specified in 4.5.4.

TABLE II. Classification of inspections.

| Title | Requirement | Inspection | First article | Conformance tests |
|---|---|---|---|---|
| Bearing seals | 3.3.1 | 4.5.5 | X | X |
| Special kits tests | 3.3.2 | 4.5.6 | X | |
| Winterization kit (M730A1) | 3.3.2.1 | 4.5.6.1 | X | |
| Cab personnel heater kit (primary) (M548A1) | 3.3.2.2 | 4.5.6.2 | X | |
| Cab personnel heater kit (secondary) (M548A1) | 3.3.2.3 | 4.5.6.3 | X | |
| Cargo area personnel heater kit (primary) (M548A1) | 3.3.2.4 | 4.5.6.4 | X | |
| Cargo area personnel heater (secondary) (M548A1) | 3.3.2.5 | 4.5.6.5 | X | |
| Engine coolant heater kit (M548A1) | 3.3.2.6 | 4.5.6.6 | X | |
| Air brake kit (M548A1) | 3.3.2.7 | 4.5.6.7 | X | |
| Material handling kit (M548A1) | 3.3.2.8 | 4.5.6.8 | X | |



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TABLE II. Classification of inspections - Continued.

| Title | Requirement | Inspection | First article | Conformance tests |
|---|---|---|---|---|
| Machine gun kit (M548A1) | 3.3.2.9 | 4.5.6.9 | X | X |
| Doors, windows, and covers | 3.3.3 | 4.5.7 | X | |
| Engine | 3.3.4 | 4.5.8 | X | X |
| Engine air induction system | 3.3.4.1 | 4.5.8.1 | X | X |
| Exhaust systems | 3.3.5 | 4.5.9 | X | |
| Fuel system, grades and slopes | 3.3.6 | 4.5.10 | X | X |
| Fuel tank | 3.3.6.1 | 4.5.10.1 | X | X |
| Fuel lines | 3.3.6.2 | 4.5.10.2 | X | X |
| Rack linkage | 3.3.6.3 | 4.5.10.3 | X | X |
| Transmission and control differential | 3.3.7 | 4.5.11 | X | X |
| Generating system voltage | 3.3.8.1 | 4.5.12.1 | X | X |
| Lights | 3.3.8.2 | 4.5.12.2 | X | X |
| Communication and electrical circuits | 3.3.8.3 | 4.5.12.3 | X | X |
| System voltage characteristics | 3.3.8.4 | 4.5.12.4 | X | |
| Electromagnetic interference suppression | 3.3.8.5 | 4.5.12.5 | X | |
| Heaters | 3.3.8.6 | 4.5.12.6 | X | |
| Controls | 3.3.9 | 4.5.13 | X | X |
| Adjustment mechanisms | 3.3.10 | 4.5.14 | X | X |
| Stowed material | 3.3.11 | 4.5.15 | X | X |
| Folding blast covers (M730A1) | 3.3.12 | 4.5.16 | X | |
| Tailgate (M73OA1) | 3.3.13 | 4.5.17 | X | |
| Air transportability | 3.3.14 | 4.5.18 | X | |
| Break-in run | 3.4 | 4.5.3 | X | |
| Reliability | 3.5.1 | 4.5.4 | X | |
| Durability | 3.5.2 | 4.5.4 | X | |
| Maintainability | 3.5.3 | 4.5.4 | X | |
| Environmental conditions | 3.5.4 | 4.5.19 | X | |
| Engine cooling | 3.5.5.1 | 4.5.20 | X | |
| Transmission lubricant cooling | 3.5.5.2 | 4.5.20 | X | |
| Differential oil cooler | 3.5.5.3 | 4.5.20 | X | |
| Bilge pumps | 3.5.6 | 4.5.21 | X | |
| Speeds and vehicle control | 3.5.7 | 4.5.22 | X | X |
| Acceleration | 3.5.8 | 4.5.23 | X | |
| Engine starting | 3.5.9 | 4.5.24 | X | X |
| Stopping and vehicle control | 3.5.10.1 | 4.5.25 | X | X |
| Holding | 3.5.10.2 | 4.5.26 | X | X 1/ |



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TABLE II. Classification of inspections - Continued.

| Title | Requirement | Inspection | First article | Conformance tests |
|---|---|---|---|---|
| Turning | 3.5.11 | 4.5.27 | X | X |
| Shallow water crossing | 3.5.12.1 | 4.5.28 | X | X |
| Amphibious operation (M730A1) | 3.5.12.2 | 4.5.29 | X | |
| Trench crossing | 3.5.13 | 4.5.30 | X | |
| Vertical obstacle | 3.5.14 | 4.5.31 | X | |
| Grade and slope operation | 3.5.15 | 4.5.32 | X | |
| Suspension | 3.5.16 | 4.5.33 | X | |
| Winch | 3.5.17 | 4.5.34 | X | |
| Winch operation | 3.5.17.1 | 4.5.34.1 | X | X |
| Winch structural integrity | 3.5.17.2 | 4.5.34.2 | X | |
| Winch controls | 3.5.17.3 | 4.5.34.3 | X | X |
| Noise level | 3.9 | 4.5.35 | X | |
| Fire extinguisher | 3.10 | 4.5.36 | X | |

1/ To be performed with vehicle unloaded.

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 (see table I) and the tests of 4.5.5, 4.5.6.9 through 4.5.8.1, 4.5.10 through 4.5.12.3, 4.5.13, 4.5.14, 4.5.22 through 4.5.28, and 4.5.34 (see table II). Prior to the inspection, the break-in shall be run (see 3.4). Following the examinations and prior to the tests, the vehicle shall be operated for a distance of eight miles. Expect for equipment and component installation, and servicing performed by the contractor as a final processing of vehicles for acceptance, vehicles shall be completely assembled and serviced with fuel, lubricants, and fluids as specified. Vehicles need not be loaded with rated payload except for holding test and amphibious operation as specified in 4.5.26 and 4.5.29.

4.4 Examination.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI Z1.4. Any redesign or modification of the contractor's standard to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements of requirements of 3.3, 3.3.1, 3.6, 3.7 and 3.8, as listed in table I. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.



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4.5 Method of inspection.

4.5.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.3, 3.3.3, 3.3.4, 3.3.6.2, 3.3.6.3, 3.3.7, 3.3.8.2, 3.3.8.3, 3.3.8.6, 3.3.9, 3.3.10, 3.5.5.1, 3.5.5.2, 3.5.5.3, 3.5.6, 3.5.16, 3.6, 3.7, 3.8, 3.10, 3.11, and 3.12 shall be determined by examination for the defects listed in table I. Examination shall be visual, tactile, or by measurement with Standard Inspection Equipment.

4.5.3 Break-in run procedure. To determine conformance to 3.4, each vehicle shall be operated on smooth, level, hard-surfaced roads as specified in table III. The undulating portion of the run shall be accomplished twice in each direction at speeds between 5 and 10 mph during division "a" and between 11 and 14.5 mph during division "b".

TABLE III. Speeds and distances for break-in run.

| Division of run | Speed - mph | Distance - miles |
|---|---|---|
| a | 0 to 10 | 5 |
| b | 11 to 20 | 5 |
| c | 21 to 30 | 10 |
| d | 31 to maximum governed | 10 |

4.5.3.1 Reverse operation. After each division of the break-in run, the vehicle shall be stopped, the engine allowed to idle for not less than one minute, and the vehicle driven in reverse for a distance of not less than 50 ft.

4.5.3.2 Oil pressure and temperature. Prior to the break-in run, the engine shall be operated at 600 to 800 rpm until lubricating oil is at operating pressure and temperature. If at any time during the break-in run the engine has been stopped for at least 15 minutes, the engine shall again be operated, as above, before continuing break-in run. Proper oil pressure and temperature shall be maintained in the engine, transmission, transfer unit, and controlled differential during the break-in run. The differential oil filter element shall be removed, cleaned, and reinstalled following completion of the break-in run.

4.5.3.3 Simulated break-in run. A simulated break-in run (see table IV) may be accomplished on the Army Skid Type Dynamometer in lieu of the run specified in 4.5.3. The final ten miles of the total break-in run shall be on smooth, level, hard-surfaced roads, except that the vehicle shall be subjected to undulating terrain by being operated over the section of road



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conforming to figure 1. The undulating portion of the test shall be accomplished twice in each direction, at speeds between 5 and 10 mph during division "e" and between 11 and 14.5 mph during division "P". After completion of division "b" and "d", table IV, the vehicle shall be operated in reverse as specified in 4.5.3.1.

TABLE IV. Simulated break-in run.

| Division of run | Speed (mph) | Miles of test | Test conditions |
|---|---|---|---|
| a | 0 to 10 | 5 | Dynamometer |
| b | 11 to 20 | 5 | Dynamometer |
| c | 21 to 30 | 5 | Dynamometer |
| d | 31 to maximum governed | 5 | Dynamometer |
| e | 21 to 30 | 5 | Track (road) |
| f | 31 to maximum governed | 5 | Track (road) |

4.5.4 5000 mile operational test. To determine conformance to 3.5.1, 3.5.2, and 3.5.3, the vehicles shall be subjected to a 5000 mile operational test as specified in table V. When vehicle is to be operated as a prime mover (equipped with air brake kit), the mileage during operation on hard-surfaced roads and cross country shall include 200 miles in each mode towing a 14 000 lb gross vehicle weight (GVW) wheeled trailer for M548A1 only.

TABLE V. 5000 mile initial production test combat loaded.

| Course | Mileage and speeds |
|---|---|
| Hard-surfaced roads 1/ | 1000 miles at varying speeds up to maximum |
| Gravel and dirt roads | 2000 miles at varying speeds up to maximum |
| Level cross country | 1000 miles at varying speeds up to maximum |
| Hilly cross country 1/ | 1000 miles at varying speeds up to maximum |
| Amphibious operation | One (1) hour minimum at varying speeds up to maximum |

1/ Hard-surfaced road and hilly cross country mileage includes 200 miles each with towed load.



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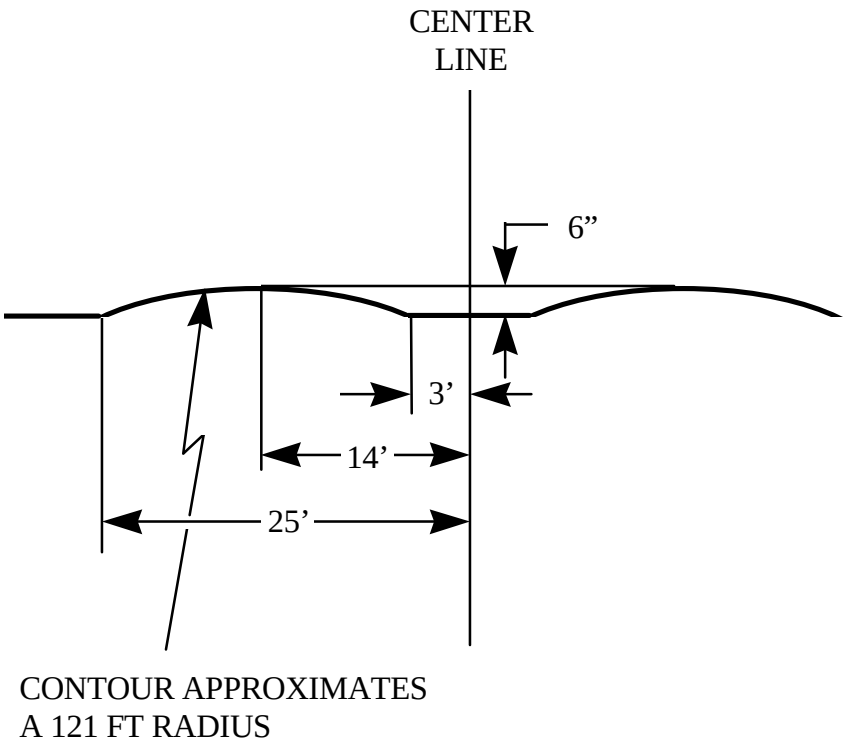



FIGURE 1. Profile of test track (artificial bump).

4.5.5 Bearing seals test. To determine conformance to 3.3.1, the vehicle shall be operated in the environments and conditions specified. Bearings and seals shall subsequently be examined for lubricant leakage and lubricant contamination.

4.5.6 Special kits examination. To determine conformance to 3.3.2, when special kits are furnished and installed, the kits shall be tested in accordance with 4.5.6 through 4.5.6.9, as applicable.

4.5.6.1 Winterization kit (M730A1). To determine conformance to 3.3.2.1, provisions for installation of winterization kit shall be examined for completeness. When specified, winterization kit shall be installed, the vehicle operated at the specified temperatures, and equipment evaluated for functional requirements.

4.5.6.2 Cab personnel heater kit (primary) (M548A1) test. To determine conformance to 3.3.2.2, the (primary) cab personnel heater kit shall be operated as in use at ambient temperatures to -25°F.



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4.5.6.3 Cab personnel heater kit (secondary) (M548A1) test. To determine conformance to 3.3.2.3, the (secondary) cab personnel heater kit shall be operated as in use at ambient temperatures from -25°F to -65°F.

4.5.6.4 Cargo area personnel heater kit (primary) (M548A1) test. To determine conformance to 3.3.2.4, the (primary) cargo area personnel heater kit shall be operated as in use at ambient temperatures to -25°F.

4.5.6.5 Cargo area personnel heater kit (secondary) (M548A1) test. To determine conformance to 3.3.2.5, the (secondary) cargo area personnel heater kit shall be operated as in use for the specified performance at ambient temperatures ranging from -25°F to -65°F.

4.5.6.6 Engine coolant heater kit (M548A1) test. To determine conformance to 3.3.2.6, the engine coolant heater kit shall be operated in ambient air temperatures between -25°F to -65°F. The engine coolant heater shall be operated on battery power alone for a minimum of twelve hours. The batteries shall then retain sufficient charge to start the engine.

4.5.6.7 Air brake kit test (M548A1). To determine conformance to 3.3.2.7, provisions for installation of the air brake kit shall be evaluated. When installed, performance shall be determined according to table II.

4.5.6.8 Material handling kit test (M548A1). To determine conformance to 3.3.2.8, the material handling kit shall load and unload material as specified. Flexibility of mounting arrangements shall also be confirmed.

4.5.6.9 Machine gun kit test (M548A1). To determine conformance to 3.3.2.9, the machine gun shall be rotated 360° to the right and left and subsequently test fired.

4.5.7 Doors, windows and covers test. To determine conformance to 3.3.3 all doors, windows and covers (including those of special kits when installed) shall be operated and examined for functional requirements. They shall be tested to assure resistance to water, dust and snow.

4.5.8 Engine test. To determine conformance to 3.3.4, the engine shall be operated throughout all speed ranges, and examined for proper installation and absence of leakage of lubricants, fuel and coolants. Prior to assembly and installation, all coolant, lubricant, and fuel lines shall be examined to insure internal cleanliness.

4.5.8.1 Engine air induction systems examination. To determine conformance to 3.3.4. 1, the engine air induction system shall be examined for proper installation and absence of leaks.



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4.5.9 Exhaust systems toxic fume test. To determine conformance to 3.3.5, the exhaust system shall be examined for proper installation and absence of leaks. Toxic fume levels measured in the driver and crew compartments, shall not exceed specified values. For the test, the vehicle shall be sealed, the engine operated at 900 to 1000 rpm and the personnel heater, if installed, operated at high heat, for a period of not less than 30 minutes.

4.5.10 Fuel system grades and slopes test. With the vehicle operating on specified grades and slopes, the vehicle engine and fuel system shall be observed to determine conformance to 3.3.6.

4.5.10.1 Fuel tank test. To determine conformance to 3.3.6.1, the fuel tank shall be examined for cleanliness. The fuel tank(s), filled to capacity, and all fuel lines and connections shall be examined for leakage before and after the testing specified in 4.5.4.

4.5.10.2 Fuel lines examination. To determine conformance to 3.3.6.2, the fuel lines shall be examined for leaks at all connections and sealed items before and after a road test on smooth, level, hard-surfaced road.

4.5.10.3 Rack linkage examination. To determine conformance to 3.3.6.3, the accelerator pedal shall be operated to full throttle, and the rack examined to assure that full open position has been reached when pedal is in contact with stop.

4.5.11 Transmission, transfer gearcase and control differential test. To determine conformance to 3.3.7, transmission and control differential shall be operated throughout all gear ranges, and examined for proper installation and absence of leaks.

4.5.12 Electrical systems test.

4.5.12.1 Generating system voltage (alternator) test, To determine conformance to 3.3.8.1, with the alternator operating as specified, a voltage measurement shall be made with a voltmeter after described load is removed, and immediately after the road test.

4.5.12.2 Lighting test. All lights shall be operated during and after break-in run throughout all stages of operation, and observed to determine conformance to 3.3.8.2.

4.5.12.3 Communication and electrical circuits test. To determine conformance to 3.3.8.3, electrical circuits supplying current to vehicle equipment shall be tested at all points to assure continuity throughout the circuits. Previously tested circuits shall be functionally tested after installation.

4.5.12.4 System voltage characteristics test. System voltage characteristics shall be evaluated as specified in MIL-STD-1275 to determine conformance to 3.3.8.4.



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4.5.12.5 Electromagnetic interference suppression test. To determine conformance to 3.3.8.5, the vehicle shall be tested as specified in MIL-STD-462.

4.5.12.6 Heaters voltage test. To determine conformance to 3.3.8.6, verify that the personnel and cargo area heaters start and operate satisfactorily when 20 Vdc is applied at the appropriate connectors.

4.5.13 Controls test. All controls shall be operated as specified and observed to determine conformance to 3.3.9.

4.5.14 Adjustment mechanisms test. To determine conformance to 3.3.10, all adjustment mechanisms shall be observed during operation, examined for proper adjustment, and adjusted if required.

4.5.15 Stowed material test. To determine conformance to 3.3.11, all manufacturer and depot installed OVM shall be stowed (combat loaded) on the vehicle. All interior and exterior items shall fit in the spaces provided. The materials shall be removed after the test. The OVM shall be of the latest configuration available to the contractor. The contractor shall be permitted to unpackage and repackage equipment used for test (for domestic shipment only).

4.5.16 Folding blast covers (M730A1). To determine conformance to 3.3.12, folding blast doors shall be observe for proper fit and function during vehicle operation.

4.5.17 Tailgate (M730A1). To determine conformance to 3.3.13, the hinged tailgate shall be observed during operation for proper installation, fit and function.

4.5.18 Air transportability (M730A1). To determine conformance to 3.3.14, examine the vehicle with the missile pallet mounted to assure the specified transportability.

4.5.19 Environmental operation test. To determine conformance to 3.5.4, vehicle shall be stored and operated at the temperatures specified.

4.5.20 Cooling system test. To determine conformance to 3.5.5.1, 3.5.5.2, and 3.5.5.3, the vehicle shall be operated at specified temperatures and cooling system temperatures measured during and after the test. Radiator pressure cap shall be pressure tested and shall relieve pressure within specified range.

4.5.21 Bilge pumps test. To determine conformance to 3.5.6, the vehicle shall have sufficient water in hull to measure bilge pumps operation. Rate of water removal, motor voltage and discharge head shall be measured.



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4.5.22 Speeds and vehicle control test. To determine conformance to 3.5.7, vehicle shall be driven at speeds from 2 to 3 mph for not less than 5 minutes and at specified speeds thereafter, and performance observed.

4.5.23 Acceleration test. To determine conformance to 3.5.8, the vehicle shall be operated as specified and acceleration measured.

4.5.24 Engine starting. To determine conformance to 3.5.9, the vehicle shall be driven up a 60% grade, engine stopped for not less than specified time and restarted. Vehicle shall then be operated on 30% side slopes and engine stopped for not less than specified time, then restarted.

4.5.25 Stopping and vehicle control test. To determine conformance to 3.5.10.1, the vehicle shall be driven at specified speed. Three stops shall be made and the results shall be averaged to determine acceptance.

4.5.26 Holding test. To determine conformance to 3.5.10.2, the vehicle shall be driven up the specified grade, brakes applied and vehicle observed for movement. Then, with brakes locked for parking, the vehicle shall again be observed for movement. During this test, vehicle shall be combat loaded, or with a simulated load of equal weight placed in a position that will not restrict the flow of engine air intake (or exhaust).

4.5.27 Turning test. To determine conformance to 3.5.11, the vehicle shall be operated as specified, turned to the right and left in full 360° turns and performance observed.

4.5.28 Shallow water fording and flotation test. To determine conformance to 3.5.12.1, the vehicle shall be driven into water of specified depth without special equipment. Water accumulation shall not exceed specified amount after 15 minutes of test.

4.5.29 Amphibious operation test (M730A1). To determine conformance to 3.3.1 and 3.5.12.2, the vehicle shall be loaded with rated payload, operated amphibiously and its performance observed. The amount of water in the hull and leakage of lubricants from the bearings shall be measured.

4.5.30 Trench crossing test. To determine conformance to 3.5.13, the vehicle shall be operated over trenches of specified depth and width.

4.5.31 Vertical obstacle test. To determine conformance to 3.5.14, the vehicle shall be operated over vertical obstacles of specified height.

4.5.32 Grade and slope operation. To determine conformance to 3.5.15, the vehicle shall be operated on specified grade and slope.



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4.5.33 Suspension test. To determine conformance to 3.5.16 vehicle, operating at a speed of 18 to 22 mph, shall cross specified obstacles without bottoming out.

4.5.34 Winch system test. To determine conformance to 3.5.17, the winch system shall be subjected to an increasing load with an 11 to 12 ft cable (total length), including three to four wraps of cable on a bare drum. The load shall be gradually increased until the shear pin fails or the load reaches 24 000 lb. The load must exceed 14 000 lb.

4.5.34.1 Winch operational test. To determine conformance to 3.5.17.1, starting with the 5th or 10th wrap on a bare cable, tension shall be applied to provide even distribution of the cable until the cable has been completely reeled in.

4.5.34.2 Winch structural integrity. To determine conformance to 3.5.17.2, the winch cable shall be pulled with a 20 000 lb load. During the test, the engine power takeoff shall be disengaged. Three to six wraps of the cable shall be wrapped on the bare drum.

4.5.34.3 Winch controls. To determine conformance to 3.5.17.3, the cab mounted winch controls shall be operated to verify engagement of the power takeoff to operate the winch. The manually operated device furnished shall be observed for free spooling of the winch.

4.5.35 Noise level. Noise levels shall be measured at operating and occasionally occupied positions while operating the equipment in its normal configuration and operational parameters (see 3.9).

4.5.36 Fire extinguisher test. To determine conformance to 3.10, the fire extinguisher system shall be examined and functionally tested (see 6.4).

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)



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6.1 Intended use. Since the carriers described in 6.1.1 and 6.1.2 must perform in battlefield conditions under which commercial items would experience catastrophic failure (in temperatures between –25 and 125°F, operation on 60% grades and 30% slopes, while fording water up to 40 in., crossing 36 in. trenches, and managing vertical obstacles up to 24 in.), these items are military unique.

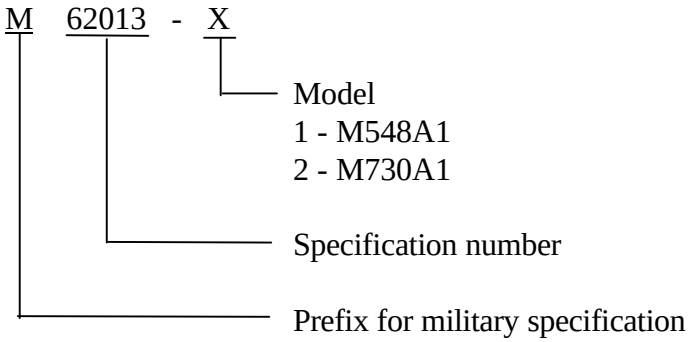
6.1.1 M548A1 carrier. The M548A1 is used as a cargo and ammunition support carrier.

6.1.2 M730A1 carrier. The M730A1 is used as a guided missile equipment carrier.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Model, PIN, and quantity of vehicles required (see 1.2 and 6.3).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. If first article samples are to be furnished and if samples are to be preproduction or initial production vehicles (see 3.1 and 4.2, 4.2.1 and 4.2.2).
e. If special exterior surface treatment is required (see 3.2.1 and 3.6).
f. If special kits for M548A1 are required (see 3.3.2).
g. If winterization kit for M730A1 is required (see 3.3.2.1).
h. If break-in run is not required (see 3.4).
i. If CARC paint is required (see 3.6).
j. If simulated break-in run is authorized (see 4.5.3.3).
k. Packaging requirements (see 5.1).

6.3 Part or identifying number (PIN). The PINs to be used for vehicles acquired to this specification are created as follows:




6.4 Safety precautions. Caution should be used in handling carbon dioxide extinguisher cylinders. Extreme care should be exercised during reinstallation to avoid tripping fire extinguisher control system since physical injury is highly probable.



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6.5 Definitions.

6.5.1 Inspection lot. An inspection lot consists of all vehicles of one model, produced during an identifiable production period, and submitted at one time for conformance inspection. Production period should not exceed one week.

6.5.2 Leakage definitions. The following definitions are used with the classification of defects for leaks:

a. Weep: Any evidence of fluid beyond the seal.
b. Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet.
c. Droplet: Any evidence of fluid beyond the seal that results in the formation of a droplet.
d. Drip: Any evidence of fluid beyond the seal where the droplet forms and falls.

6.6 International standardization agreement. Certain provisions (see 3.5.12) of this specification are the subject of international standardization agreements NATO STANAG 2805-A (Edition 3), and ABCA-ARMY-STD-151. When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

6.7 Subject term (key word) listing.

Amphibious
Cab
Diesel
Hull
Suspension
Transmission

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army – AT

Preparing Activity:
Army - AT

(Project 2350-0505)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-DTL-62013E(AT) | 2. DOCUMENT DATE *(YYMMDD)*<br>980930 |
|---|---|---|

**3. DOCUMENT TITLE**
CARRIERS, FULL-TRACKED: M548A1 AND M730A1

**4. NATURE OF CHANGE** ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS *(Include Zip Code)* | d. TELEPHONE *(Include Area Code)*<br>(1) Commercial<br>(2) AUTOVON<br>*(If applicable)* | 7. DATE SUBMITTED<br>*(YYMMDD)* |

**8. PREPARING ACTIVITY**

| a. NAME | b. TELEPHONE *(Include Area Code)*<br>(1) Commercial (810) 574-8745 (2) AUTOVON 786-8745 |
|---|---|
| c. ADDRESS *(Include Zip Code)*<br>Commander<br>U.S. Army Tank-automotive and Armaments Command<br>ATTN: AMSTA-TR-E/BLUE<br>Warren, MI 48397-5000 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403<br>Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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